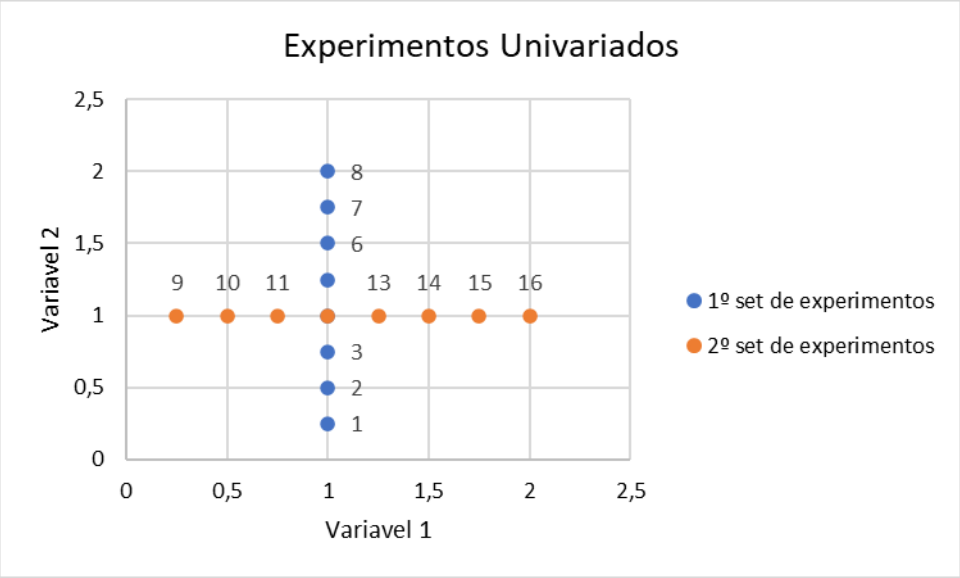
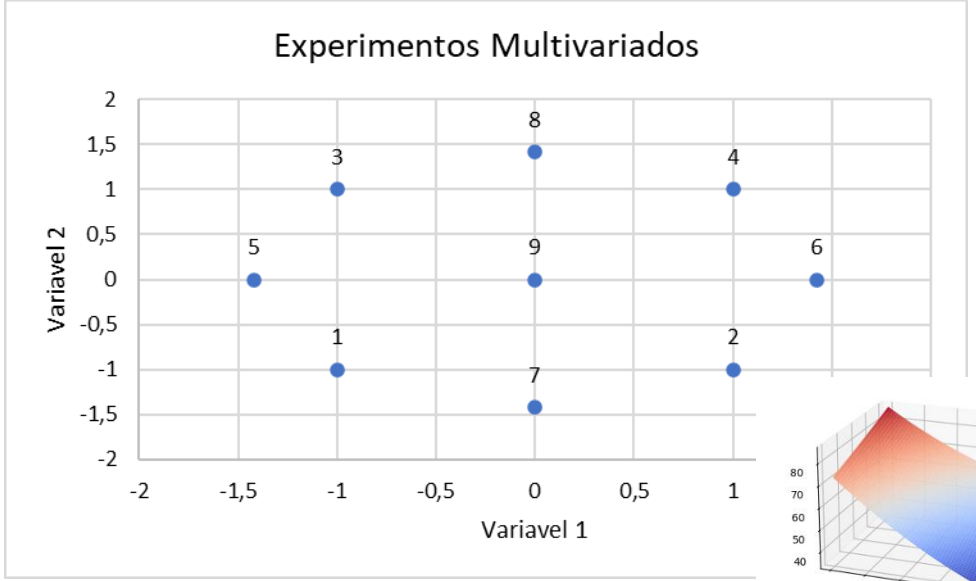
Conceito: análise univariada vs multivariada

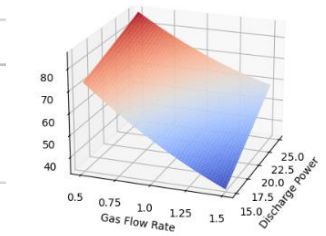
One Variable at a time: uma variável por vez



- O experimentalista fixa a variável 1 em 1un e varia a variável 2 até encontrar a melhor resposta. Depois fixa a variável 2 na melhor condição encontrada e varia a variável 1 até encontrar a melhor condição.
- A interação entre as variável é ignorada.
- Não há informação sobre como a resposta se comporta em grande parte do domínio experimental.
- Grande quantidade de experimentos.

Response surface method: método da superfície de resposta

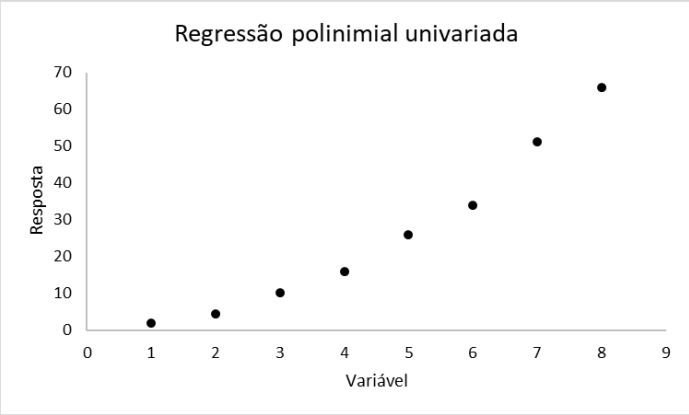




- As variáveis são variadas simultaneamente.
- Todo o domínio experimental é investigado.
- A interação entre as variáveis é investigada.
- A resposta estudada é equacionada e pode ser prevista para cada valor das variáveis independentes.
- Menor quantidade de experimentos.

Conceito: método de regressão.

Regressão univariada

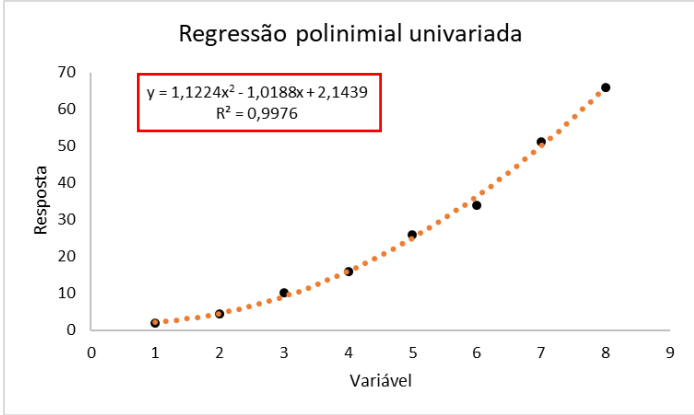


Matriz X = condições da variável independente

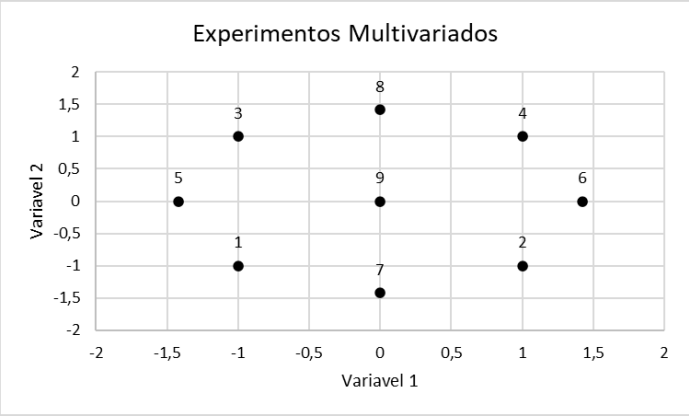
Vetor y = respostas obtidas

$$b = (X^t X)^{-1} X^t y$$

Método dos mínimos múltiplos quadráticos
b = coeficientes de regressão



A regressão multivariada é realizada de forma análoga, para duas ou mais variáveis independentes



Matriz X = condições da variável independente

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$$b = (X^t X)^{-1} X^t y$$

Método dos mínimos múltiplos quadráticos
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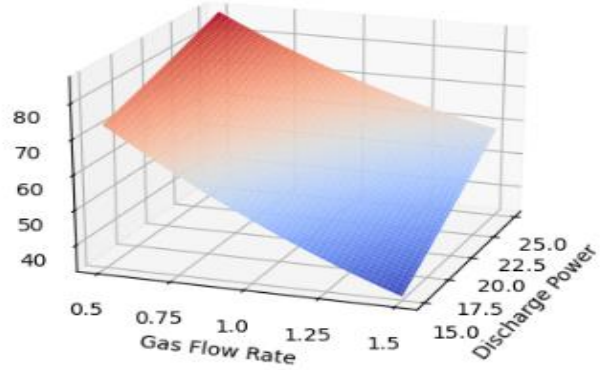

Algoritmo

Inputs:
Matriz X
Vetor Y

Coeficientes de regressão

$$b = (X^t X)^{-1} X^t y$$

Resultados previstos

$$\hat{y} = Xb$$

$$\hat{y} - y = res$$

- SQR
- Sqres
- SQT

Intervalo de confiança dos coeficientes (IC)

Exclusão dos coeficientes insignificantes

Modelo de regressão completo

$$Y_1 = 59.69 + 4.75x_1 - 8.43x_2 - 4.99x_3 + 0.74x_1x_2 - 0.67x_1x_3 + 1.86x_2x_3 + 0.075x_1^2 + 0.72x_2^2 + 0.79x_3^2$$

Referência 1

Chemosphere 155 (2016) 9–17

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journal homepage: www.elsevier.com/locate/chemosphere

Investigation of hybrid plasma-catalytic removal of acetone over CuO/γ-Al_2O_3 catalysts using response surface method

Xinbo Zhu [a], Xin Tu [b, *], Danhua Mei [b], Chenghang Zheng [a], Jinsong Zhou [a], Xiang Gao [a, **], Zhongyang Luo [a], Mingjiang Ni [a], Kefa Cen [a]

[a] State Key Laboratory of Clean Energy Utilization, Zhejiang University, Hangzhou 310027, PR China
[b] Department of Electrical Engineering and Electronics, University of Liverpool, Liverpool L69 3GJ, UK

HIGHLIGHTS

- Acetone removal was improved using CuO/γ-Al_2O_3 catalysts in a plasma reactor.
- 5 wt% CuO/γ-Al_2O_3 exhibited the best plasma-catalytic performance for the removal of acetone.
- Response surface method was used to evaluate the importance of different operating parameters.
- The gas flow rate was the most significant factor to determine the removal efficiency of acetone.
- The initial concentration of acetone played the most important role in the energy efficiency.

Resumo do paper

Objetivo:
Otimizar a purificação do ar (remoção de acetona)
Menor gasto energético possível.

Respostas:
Eficiência de remoção (%)
Eficiência energética (g/kWh)

Meio:
Sistema de fechado de fluxo de ar controlado, onde a purificação é realizada por meio de descarga elétrica.
Para maiores detalhes consultar paper

Problema:
Encontrar as melhores condições de fluxo de ár, intensidade da descarga elétrica e concentração inicial de acetona no sistema.

Variáveis:
Intensidade de descarga (x_1)
Fluxo de ar (x_2)
Concentração inicial de acetona (x_3)

Método:
Regressão multivariada – Response Surface Method (RSM)

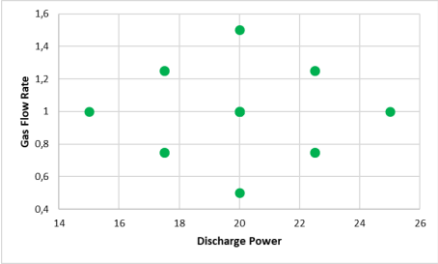
Esquema

Definição das variáveis e respostas avaliadas

Variáveis
- Discharge Power (15 – 22,5W)
- Gas Flow Rate (0,5 – 1,5 L/min)
- Initial concentration of Accetone

Respostas
- Removal Efficiency (%)
- Energy Efficiency (g/kWh)

Organização do planejamento experimental.
Realização dos experimentos e obtenção das respostas em cada condição estabelecida



X0	X1	X2	X3	X12	X13	X23	X11	X22	X33	y1	y2
1	20	1	200	20	4000	200	400	1	40000	59,3	0,91
1	20	1	200	20	4000	200	400	1	40000	59,2	0,91
1	20	0,5	200	10	4000	100	400	0,25	40000	78,5	0,6
1	15	1	200	15	3000	200	225	1	40000	49,7	1,02
1	20	1	200	20	4000	200	400	1	40000	59,5	0,92
1	17,5	0,75	250	13,125	4375	187,5	306,25	0,5625	62500	59,7	0,98
1	20	1	300	20	6000	300	400	1	90000	51,1	1,18
1	22,5	0,75	250	16,875	5625	187,5	506,25	0,5625	62500	68,7	0,88
1	22,5	1,25	250	28,125	5625	312,5	506,25	1,5625	62500	55,4	1,18
1	20	1	200	20	4000	200	400	1	40000	59,3	0,91
1	17,5	1,25	250	21,875	4375	312,5	306,25	1,5625	62500	47,2	1,3
1	25	1	200	25	5000	200	625	1	40000	67,9	0,84
1	22,5	0,75	150	16,875	3375	112,5	506,25	0,5625	22500	81,3	0,63
1	20	1,5	200	30	4000	300	400	2,25	40000	44,3	1,02
1	20	1	100	20	2000	100	400	1	10000	72,2	0,56
1	22,5	1,25	150	28,125	3375	187,5	506,25	1,5625	22500	64,3	0,82
1	20	1	200	20	4000	200	400	1	40000	59,2	0,91
1	17,5	0,75	150	13,125	2625	112,5	306,25	0,5625	22500	73,4	0,73
1	20	1	200	20	4000	200	400	1	40000	59,3	0,91
1	17,5	1,25	150	21,875	2625	187,5	306,25	1,5625	22500	49,7	0,82

Regressão e validação estatística do modelo

Modelo de regressão e avaliação das condições experimentais

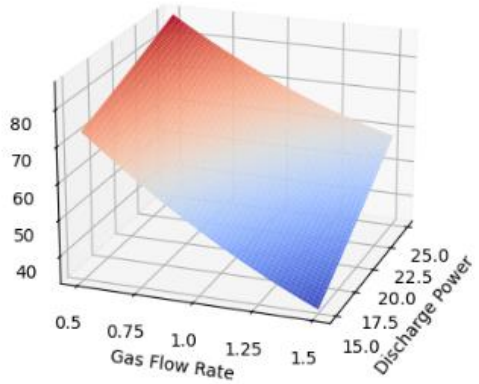
Removal Efficiency(%):

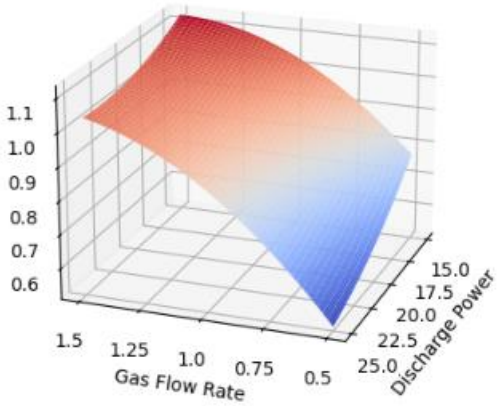
$$Y_1 = 59.69 + 4.75x_1 - 8.43x_2 - 4.99x_3 + 0.74x_1x_2 - 0.67x_1x_3 + 1.86x_2x_3 + 0.075x_1^2 + 0.72x_2^2 + 0.79x_3^2 \quad (5)$$

Energy Efficiency $(g \cdot kWh^{-1})$:

$$Y_2 = 0.92 - 0.043x_1 + 0.11x_2 + 0.16x_3 + 0.012x_1x_2 - 0.015x_1x_3 + 0.04x_2x_3 + 7.222 \times 10^{-3}x_1^2 - 0.021x_2^2 - 7.736 \times 10^{-3}x_3^2 \quad (6)$$

Removal Efficiency (%)





Resultados

	coef	var	err	IC	|coef| - |IC|
X0	59.695455	0.475775	0.689764	1.560355	58.135099
X1	4.756250	0.186912	0.432333	0.978005	3.778245
X2	-8.431250	0.186912	0.432333	0.978005	7.453245
X3	-4.993750	0.186912	0.432333	0.978005	4.015745
~~X12~~	~~0.737500~~	~~0.373823~~	~~0.611411~~	~~1.383107~~	~~-0.645607~~
~~X13~~	~~-0.662500~~	~~0.373823~~	~~0.611411~~	~~1.383107~~	~~-0.720607~~
X23	1.862500	0.373823	0.611411	1.383107	0.479393
~~X11~~	~~0.072727~~	~~0.118944~~	~~0.344882~~	~~0.780178~~	~~-0.707450~~
~~X22~~	~~0.722727~~	~~0.118944~~	~~0.344882~~	~~0.780178~~	~~-0.057450~~
X33	0.785227	0.118944	0.344882	0.780178	0.005050

$$remoção = 59{,}67 + 4{,}75x_1 - 8{,}43x_2 - 5{,}00x_3 + 0{,}73x_{12} - 0{,}66x_{13} + 1{,}86x_{23} + 0{,}72x_1x_1 + 0{,}72x_2x_2 + 0{,}78x_3x_3$$

$$\Rightarrow remoção = 59{,}67 + 4{,}75x_1 - 8{,}43x_2 - 5{,}00x_3 + 1{,}86x_{23} + 0{,}78x_3x_3$$

Removal Efficiency(%):

$$Y_1 = 59.69 + 4.75x_1 - 8.43x_2 - 4.99x_3 + 0.74x_1x_2 - 0.67x_1x_3 + 1.86x_2x_3 + 0.075x_1^2 + 0.72x_2^2 + 0.79x_3^2 \quad (5)$$

Modelo de regressão obtido pelos autores

	coef	var	err	IC	|coef| - |IC|
X0	0.918409	0.000141	0.011855	0.026818	0.891591
X1	-0.042500	0.000055	0.007431	0.016809	0.025691
X2	0.108750	0.000055	0.007431	0.016809	0.091941
X3	0.161250	0.000055	0.007431	0.016809	0.144441
~~X12~~	~~0.010000~~	~~0.000110~~	~~0.010508~~	~~0.023772~~	~~-0.013772~~
~~X13~~	~~-0.015000~~	~~0.000110~~	~~0.010508~~	~~0.023772~~	~~-0.008772~~
X23	0.042500	0.000110	0.010508	0.023772	0.018728
~~X11~~	~~0.007955~~	~~0.000035~~	~~0.005928~~	~~0.013409~~	~~-0.005454~~
X22	-0.022045	0.000035	0.005928	0.013409	0.008636
~~X33~~	~~-0.007045~~	~~0.000035~~	~~0.005928~~	~~0.013409~~	~~-0.006364~~

$$eficiência\ energética = 0{,}92 - 0{,}04x_1 + 0{,}11x_2 + 0{,}16x_3 + 0{,}01x_{12} - 0{,}01x_{13} + 0{,}04x_{23} + 0{,}01x_1x_1 - 0{,}02x_2x_2 - 0{,}01x_3x_3$$

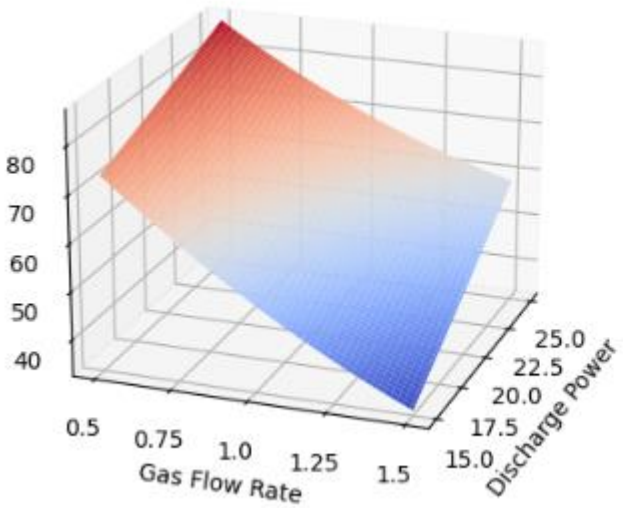
$$\Rightarrow eficiência\ energética = 0{,}92 - 0{,}04x_1 + 0{,}11x_2 + 0{,}16x_3 + 0{,}04x_{23} - 0{,}02x_2x_2$$

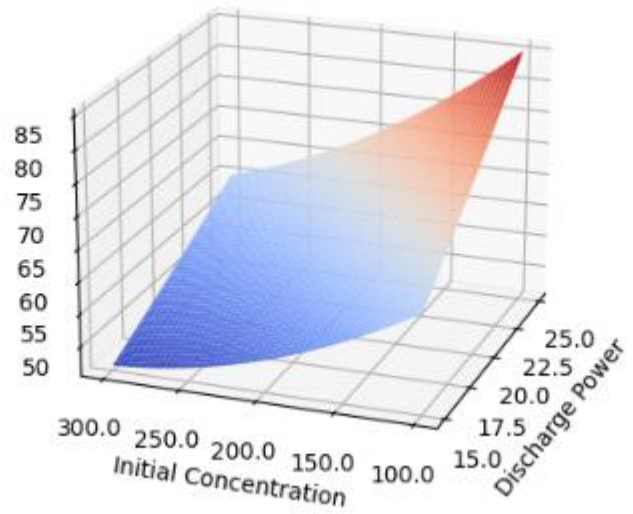
Energy Efficiency $(g \cdot kWh^{-1})$:

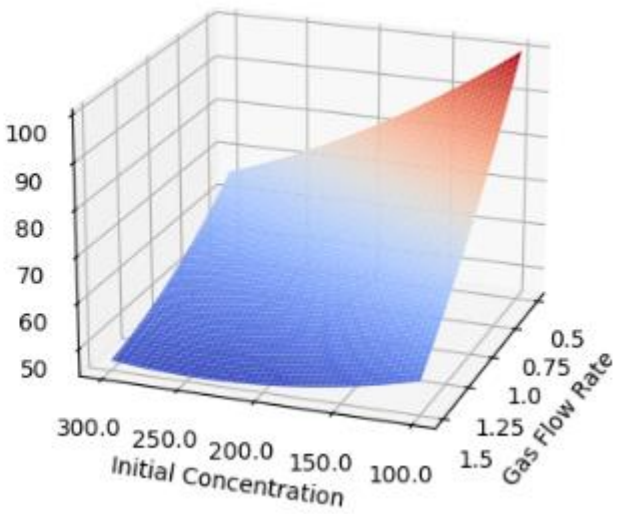
$$Y_2 = 0.92 - 0.043x_1 + 0.11x_2 + 0.16x_3 + 0.012x_1x_2 - 0.015x_1x_3 + 0.04x_2x_3 + 7.222 \times 10^{-3}x_1^2 - 0.021x_2^2 - 7.736 \times 10^{-3}x_3^2 \quad (6)$$

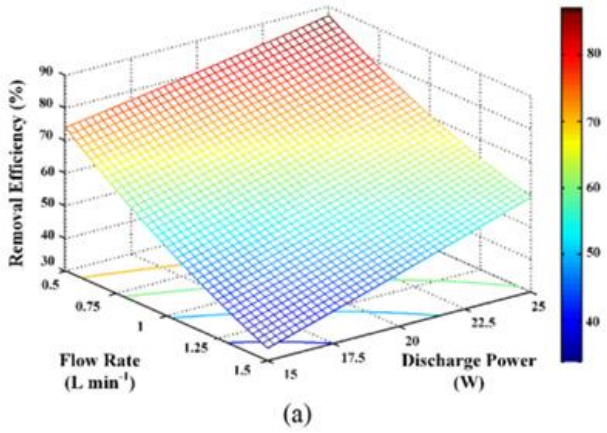
Modelo de regressão obtido pelos autores

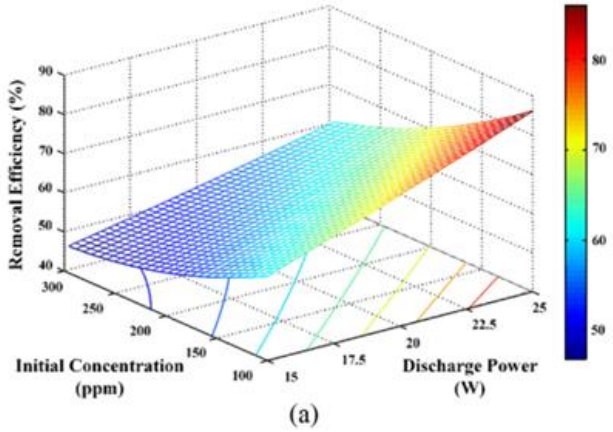
Resultados: Superfícies de Resposta para a Eficiência de Remoção

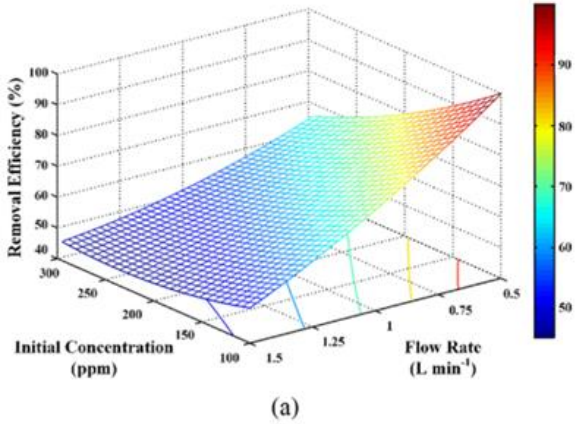






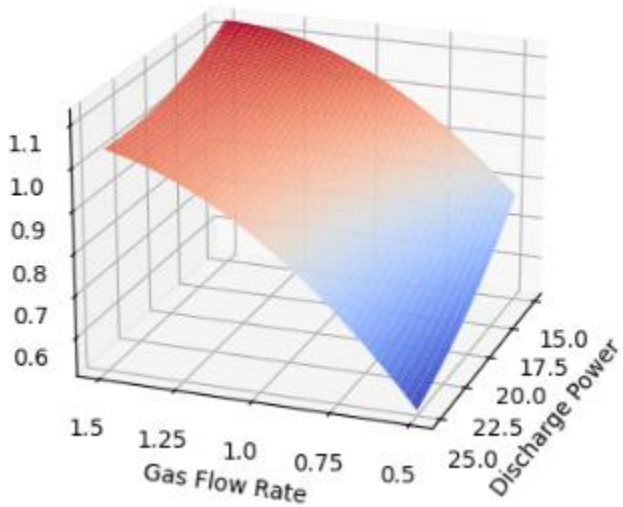


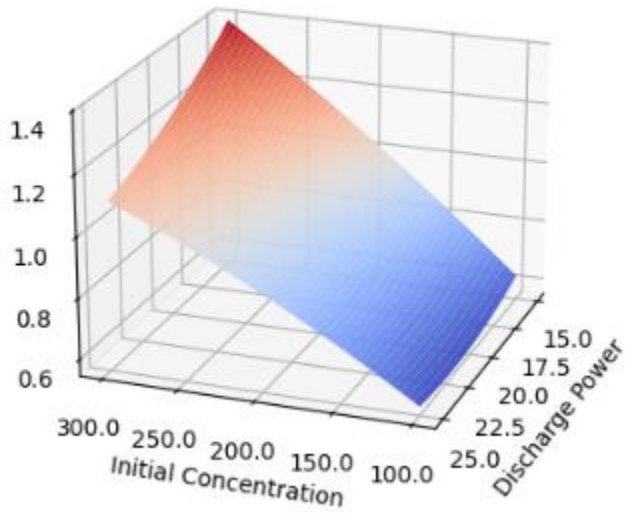


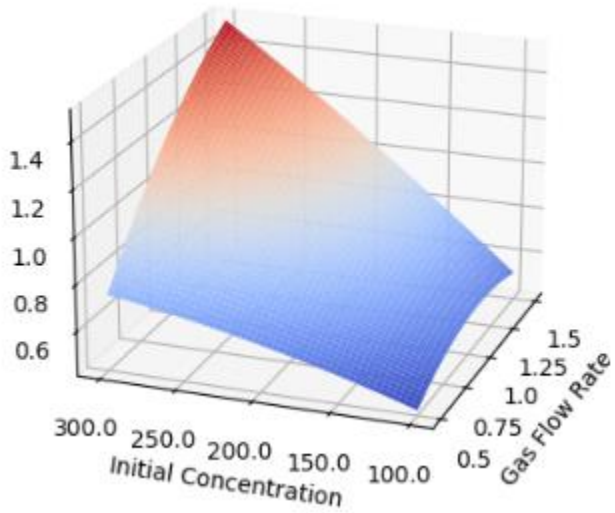


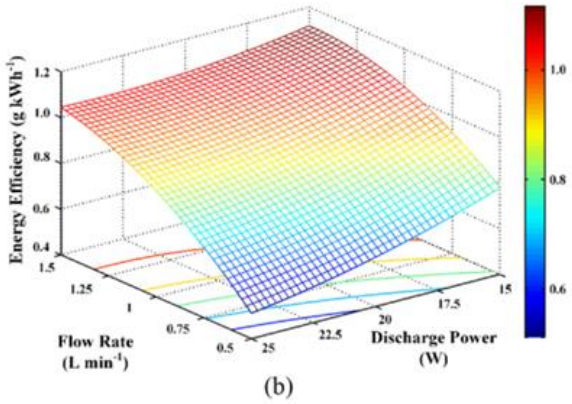
Superfícies obtidas pelos autores

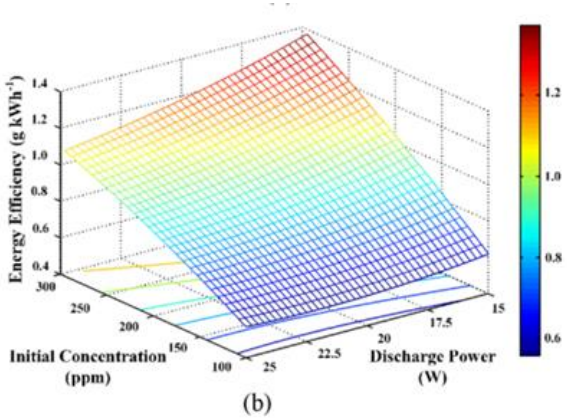
Resultados: Superfícies de Resposta para a Eficiência Energética

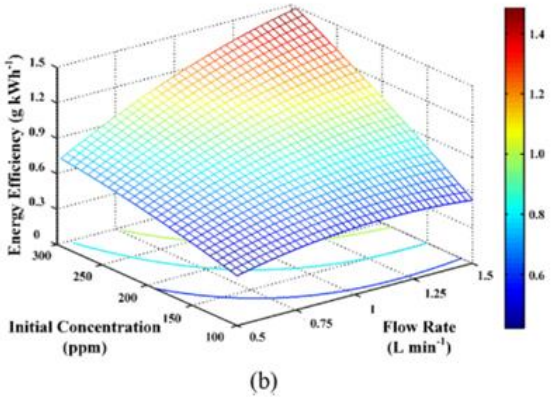












Superfícies obtidas pelos autores